AACNet: Asymmetric Attention Convolution Network for Hyperspectral Image Dehazing

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***Abstract*— Haze in hyperspectral images (HSIs) can lead to crosstalk between multiple bands, resulting in errors that can be amplified and transmitted during data processing. As a consequence, this may cause a reduction in the accuracy and precision of remote sensing data. The purpose of haze removal is to restore high-quality HSIs from degraded ones. The high spectral resolution and typically dozens to hundreds of spectral bands in HSIs pose significant challenges for haze removal. Thus, many methods designed for natural and multispectral images are not effective in removing haze in HSIs. To address this challenge, we develop a model called asymmetric attention convolution network (AACNet) designed for haze removal in HSIs. Specifically, the basic architecture of AACNet is mainly composed of several residual asymmetric attention groups (RAAGs), where the core components are residual asymmetric attention blocks (RAABs). This design enables the full utilization of deep spatial–spectral features while skipping low-frequency regions and focusing more on the haze-affected areas. To more accurately restore the spectral information in areas polluted by haze, a pooling channel self-attention (PCSA) module has been proposed. This module can effectively reconstruct the spectral response curve that is affected by the haze. Our experiments on both simulated and real datasets demonstrate that the proposed AACNet outperforms several leading haze removal methods in both precision and visual quality. The source code and data of this article will be made publicly available at https://github.com/SZU710/AACNet for reproducible research.**

***Index Terms*— Attention mechanism, deep learning, haze removal, hyperspectral images (HSIs).**

I. Introduction

HYPERSPECTRAL sensors can image objects in continuous narrow wavelengths, producing spectral curves that reflect the object's physical properties. Due to the unique spectral characteristics of ground objects, hyperspectral images (HSIs) can accurately detect, identify, and analyze ground objects. HSIs are widely used in Earth observation applications, such as resource exploration [1], [2], precision agriculture [3], [4], and disaster monitoring [5], [6], [7]. Optical imaging is susceptible to scattering and atmospheric particulate matter, resulting in blurry images, reduced visibility, and low contrast, all of which significantly degrade image fidelity. Haze, in particular, can destroy the spectral curve of surface objects and seriously affect the accurate identification of HSIs, ultimately limiting its practical applications. As a result, image dehazing is crucial for many image-processing applications and holds significant practical value.

Researchers have proposed various methods for restoring clear images from hazy images, but these methods are typically specific to natural images. In general, there are two categories of image dehazing methods: prior-based and learning-based. The most striking difference between these two types is that the image priors in prior-based methods are manually crafted, while learning-based methods automatically learn the priors. Several typical image priors, such as dark channel prior [8], color-line prior [9], and color attenuation prior [10] have been proposed to calculate the transmission map for achieving clear images. In addition, Berman et al. [11] discovered that image dehazing can also be accomplished through a nonlocal haze-line prior. Although these prior-based methods can produce satisfactory results in some specific scenarios, the reliance on prior assumptions may lead to inaccurate dehazing results. It is worth noting that the atmospheric scattering model-based RGB image dehazing methods [12], [13] are not suitable for haze removal in HSIs. This is mainly because the atmospheric scattering model assumes that the attenuation coefficient of all spectral channels is constant, which does not hold for HSIs due to the more extensive spectral information they possess.

In recent years, deep learning-based methods have demonstrated significant advantages in computer vision and image processing. With the expansion of image datasets, deep learning has become increasingly popular in the study of removing haze from RGB images. Ren et al. [12] and Cai et al. [13] utilized convolutional neural networks (CNNs) to estimate the medium transmission map from the blurry input image, which was then used to generate the corresponding haze-free image. However, the inaccurate estimation of the transmission map always resulted in a decrease in the quality of the dehazed image. Therefore, subsequent researchers proposed a series of dehazing methods that do not require the estimation of the medium transmission map. Chen et al. [14] employed smooth dilation convolution to avoid grid artifacts, and adopted a gating subnetwork to fuse features from different levels, thereby recovering clear haze-free images. With the development of

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attention mechanisms in deep learning, researchers began to use attention mechanisms to enhance the representation ability of dehazing networks. Qin et al. [15] combined channel and spatial attention mechanisms to selectively process different channel and pixel weights, enabling the network to focus more on the contaminated areas. Song et al. [16] introduced gating mechanisms into convolutional blocks to serve as pixel attention modules, thereby enhancing the dehazing performance of the model.

In the field of remote sensing, a large number of models for multispectral image dehazing have been proposed. These models can be mainly divided into two categories: dehazing models based on traditional methods and dehazing models based on deep learning. The traditional model-based or prior-based methods rely on manually extracted features from the hazy images and use various mathematical models to estimate and remove the haze. Zhou and Wang [17] utilized radiative transfer model to remove thin clouds by calculating the linear regression coefficient between visible or infrared bands and cirrus band. Xie et al. [18] first determined the hazy area through the image brightness, and then repaired the image based on color consistency constraints and global prior information. Xu et al. [19] developed a dehazing method that employs a dehazing operator to gradually and iteratively remove haze until satisfactory results are obtained, and updates the transmission rate and scene brightness information of the image through multiple iterations before reconstructing the final clear image using the estimated transmission rate and the hazy scene. Xu et al. [20] found that clouds have a higher signal-to-noise ratio, and they removed thin clouds in optical remote sensing images through noise-adjusted principal component transformation. These methods have been widely used and achieved some success in dehazing. However, they have limitations in dealing with complex and diverse hazy scenes, and low accuracy and robustness in the restoration of remote sensing images due to their reliance on handcrafted features and assumptions.

With the development of deep learning technology, researchers have begun to use deep learning techniques to solve the problem of multispectral image dehazing. CNNs with residual structures were applied to increase the depth of the model and improve its ability to extract haze features [21], effectively enhancing the dehazing performance of the model. Guo et al. [22] integrated local and global residual learning into the network, and introduced a channel attention module to capture strong channel correlations present in multispectral images. Zi et al. [23] employed a neural network designed based on the U-Net [24] architecture to estimate the thickness coefficients of each band with respect to the reference haze thickness map, thereby obtaining haze maps of different bands. The approach proposed in [25] leverages haze distortion control functions and a multiscale grid architecture to intelligently model and restore haze-free images while preserving the spatial structures of ground objects. These techniques aim to alleviate the degradation caused by haze in multispectral remote sensing images. Nevertheless, the straightforward application of these techniques to haze removal in HSIs is a formidable task, mainly due to the augmented spectral complexity inherent in HSIs.

The haze present in remote sensing images is spatially variant and its effect is related to the wavelength, with a tendency to decrease as the wavelength increases [26]. This phenomenon causes color distortion in the image and affects the performance of traditional dehazing methods that are based on hand-crafted features. The impact of haze in HSIs is more complex than that in multispectral images. Different bands are affected by haze to varying degrees, making it more challenging to remove haze from all bands compared to multispectral images. Furthermore, the bands in HSIs are highly correlated, which means that there is a significant overlap of information between them, making it easier to overfit the model. Moreover, HSIs typically contain hundreds of bands, requiring more feature information to reconstruct a clear HSI during the haze removal process than in multispectral images. This necessitates the model to have stronger spatial and spectral feature extraction capabilities to achieve better haze removal results. Therefore, applying the multispectral image haze removal method directly to HSIs may lead to suboptimal results. To address these challenges, we propose a new HSI haze removal method called asymmetric attention convolutional network (AACNet). Our proposed AACNet can extract comprehensive and complex spatial and spectral features, capture the correlations between different channels to prevent overfitting, and finely adjust the weights of different channels, effectively reconstructing the spectral response curves of the haze-contaminated regions. The three main contributions of this work are described as follows.

1) The backbone network architecture of AACNet is mainly composed of several residual asymmetric attention groups (RAAGs), with residual asymmetric attention blocks (RAABs) being the core component of each RRAG. The proposed RAAB adopts a multipath connection to fully utilize the deep spatial–spectral features and uses residual connections and attention mechanisms to skip low-frequency regions and focus on high-frequency regions.
2) In RAAG, asymmetric attention convolution (AAConv) is the most critical component, which consists of fusion asymmetry convolution (FAConv) and global attention (GA). During the training phase, the proposed FAConv utilizes multipath connections to fully exploit the deep spatial–spectral features. In the deployment phase, by employing the structural reparameterization strategy, FAConv can be precisely and losslessly converted to a memory-friendly single-path convolution, which improves the model's running speed while ensuring the same performance. GA enables AAConv to pay more attention to the hazy regions of HSIs.
3) To more accurately restore the spectral information of regions polluted by haze, a pooling channel self-attention (PCSA) module is designed to capture the correlations between different channels and finely adjust the weights of different channels. The developed PCSA can effectively reconstruct the spectral response curve affected by pollution. The proposed method has been evaluated on various simulated and real datasets, and the experimental results show that AACNet outperforms

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the state-of-the-art dehazing methods in terms of both qualitative and quantitative evaluation metrics.

The rest of this article is organized below. Section II introduces related work on HSI dehazing, attention mechanism, and structural reparameterization. Section III details the proposed approach. Section IV validates the effectiveness of our approach on both simulated and real datasets. Finally, Section V concludes this article.

II. Related Works

A. Dehazing Models for HSIs

There is limited literature on dehazing for HSIs. Furthermore, the majority of these studies have relied on model-driven approaches. Kang et al. [27] first calculated a haze density map by differentiating the average bands belonging to the visible and infrared spectral ranges. The haze intensity map mainly reflects the spatial distribution of haze in the hazy image. Then, the haze abundance and density in different spectral bands were estimated based on the pixel reflectance between two manually selected pixels with different haze levels. Finally, the haze-free HSIs were obtained by subtracting the product of the estimated haze abundance and density maps from the original hazy target image. To automatically estimate the concentration of haze, Li et al. [28] proposed a saliency measure method that utilizes the reflectance differences between hazy and haze-free pixels. The RGB bands are extracted from the original HSI and transformed into the intensity-hue-saturation (IHS) space. Then, the saliency maps for haze-free and hazy pixels are computed. Ultimately, the haze intensity map is integrated with the saliency maps to estimate the reflectance of hazy and haze-free pixels.

Another work based on a spectral grouping network (SGNet) was proposed by Ma et al. [29] by utilizing the useful information of each spectral band. SGNet divided HSIs into spectral subsets based on their correlation and performed parallel convolution with multiple branches for feature extraction. To transfer useful information between adjacent branches, a novel attention block was designed. Despite the effectiveness of SGNet, its spectral grouping strategy may compromise the continuity of spectral bands, leading to reduced accuracy in reconstructing HSIs. To overcome these limitations, we propose AACNet, a model that can be applied to haze removal from HSIs with various numbers of bands without grouping, while preserving the continuity of the spectra. AACNet consists of multiple RRAGs, where RAAB is the core component of each RRAG. The proposed RAAB utilizes a multipath connection to fully leverage deep spatial–spectral features and employs residual connections and attention mechanisms to skip low-frequency regions and focus on high-frequency regions. Moreover, AACNet can preserve the original features of the HSI through skip connections.

B. Attention Mechanism

The attention mechanism has gained significant attention in remote sensing image processing [30], [31], [32], [33] due to its ability to selectively emphasize informative features and suppress irrelevant ones, inspired by human perceptual mechanisms. Previous research has explored the effectiveness of attention mechanisms in various remote sensing applications, such as HSI denoising [34], change detection [35], image classification [36], and band selection [37]. However, to the best of our knowledge, attention mechanisms have not been applied to the HSI dehazing problem. In this article, we present the PCSA block, which aims to accurately reconstruct the spectra of HSIs contaminated by haze. The PCSA block uses a spatial pooling operation to obtain a channel descriptor, assigns a weight to each channel, and calculates the self-attention matrix of the channel descriptor to finely adjust the weights of different channels and effectively reconstruct the spectra contaminated by haze.

C. Structural Reparameterization

Structural reparameterization is a technique that parameterizes a structure using parameters transformed from another structure [38], [39]. Although this approach may result in heavy parameters, large memory consumption, and slow running speed during model training, it can ensure the model has strong representation capabilities. Using the structural reparameterization technique during deployment, the model can inherit powerful representation capabilities and achieve a fast running speed. Various such approaches have been proposed to improve the performance of CNNs. ExpandNets [40] expanded the traditional convolutional layers into three consecutive convolutional layers, preserving the advantages of different sensory field size convolutions to enhance model performance. Reparameterization visual geometry group (ReVGG) network [41] added a multibranch topology to the VGG model, which effectively improved its performance [42]. Asymmetric convolutional network (ACNet) [43] enhanced the robustness of traditional convolution against rotational distortions using 1-D asymmetric convolution. During deployment, ACNet is equivalent to a traditional convolution. To enhance the capability of exploring the deep spatial–spectral features of HSIs, we propose FAConv. During the model training phase, FAConv utilizes a multibranch structure to aggregate features extracted from different branches, effectively improving the model's ability to extract spatial and spectral features. In the deployment phase, FAConv can be mathematically converted into a single 3×3 convolution without compromising accuracy, leading to a faster runtime.

III. Proposed Method

A. Architecture of AACNet

Let $\mathcal{X} \in \mathbb{R}^{H\times W\times C}$ represent the hazy HSI and $\mathcal{Y} \in \mathbb{R}^{H\times W\times C}$ represent the clear HSI. Here, C is the number of channels, and H and W are the height and width of the image, respectively. We assume that $\mathcal{X}$ and $\mathcal{Y}$ satisfy the following relationship:

$$\mathcal{X} = \mathcal{Y}t + a(1-t) \tag{1}$$

where a is the global atmospheric light and t is the haze transmission diagram. Thus, a clear HSI can be obtained by the following equation:

$$\mathcal{Y} = \frac{\mathcal{X} - a}{t} + a. \tag{2}$$

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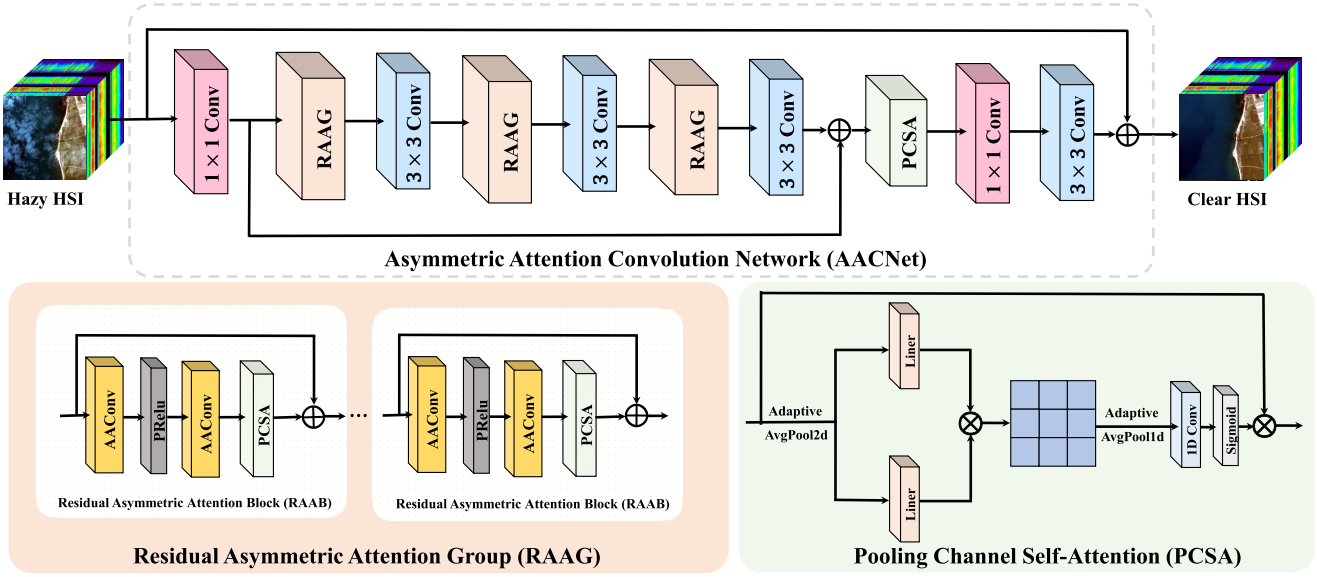


Fig. 1. Network architecture of the proposed AACNet. ⊕ denotes the elementwise summation. ⊗ denotes the elementwise product.

In hyperspectral imaging, each pixel is represented by a spectrum of intensities at different wavelengths, and the presence of haze in the atmosphere causes the spectral intensities to be attenuated in a wavelength-dependent manner. The amount of attenuation is determined by a and t, both of which are unknown and need to be estimated to perform dehazing. Thus, dehazing from HSIs is an ill-posed problem. To address this issue, we propose a deep learning model called AACNet which is designed to learn deeper and more diverse spatial and spectral information, enabling it to capture a wider range of features compared to models that only focus on one aspect. Furthermore, AACNet can finely adjust the weights of different channels, which enhances its ability to efficiently predict clear HSIs from haze ones.

As shown in Fig. 1, AACNet uses a combination of jump connections, convolutional layers, and attention mechanisms to perform haze removal on HSIs. The jump connections are used to preserve the original features of the hazy HSI, which can be important for maintaining the overall structure and content of the image. The 1×1 convolutional layer is then used to perform shallow feature extraction, which can help to identify more specific features related to the presence of haze. The three RAAGs and 3×3 convolutional layers connected in series can help to further extract features from the input image, with the RAAGs allowing the model to selectively attend to different parts of the input based on their relevance to the task. The proposed PCSA, 1×1 convolutional layer, and 3×3 convolutional layer are used to further extract and refine the features obtained from the RAAGs. The resulting features are then added to the shallow features obtained previously, allowing the model to take into account both shallow and deeper features in the image. Finally, the resulting features are added to the original features of the HSI to obtain a clear image.

The RAAG in AACNet is composed of n RAABs, each of which consists of local residual learning, AAConv, parametric rectified linear unit (PReLU), and PCSA. The RAAB utilizes local residual learning to bypass the hazy regions and preserve the regions that are not contaminated by haze. PRelu activation function can help to introduce nonlinearity into the model and improve its ability to capture complex relationships between the input features. The combination of AAConv and PCSA can fully explore the spatial and spectral features of the hazy regions and help the model better understand the distribution and characteristics of the haze in the image ultimately leading to more effective haze removal for HSIs.

B. AAConv

As shown in Fig. 2, during the training phase AAConv consists of two parts: FAConv and GA. The input features are both fed into FAConv and GA, and the final output features are obtained by cross-multiplying them. In FAConv, we parallelized 3×3 convolution, 1×3 convolution, 3×1 convolution, and 1×1 convolution. This parallel design architecture allows FAConv to learn deeper and more diverse spatial and spectral information compared to ordinary convolutions. When a network has multiple parallel branches, each branch increases the memory consumption of the network, and this may increase the model's running time and deployment costs. Fortunately, using structural reparameterization techniques, a complex multibranch network can be transformed into a simple single-branch network without loss during the model deployment phase. Fig. 3 illustrates how the deployment phase FAConv converts the multibranch training phase FAConv in Fig. 2 into an equivalent single-branch 3×3 convolution process. Once the training phase FAConv in Fig. 2 is trained, the parameters of each convolution are determined. Then, in the structural reparameterization process shown in Fig. 3, the parameters of 3×3 convolution, 1×3 convolution, 3×1 convolution, and 1×1 convolution are added together and assigned to the 3×3 convolution, resulting

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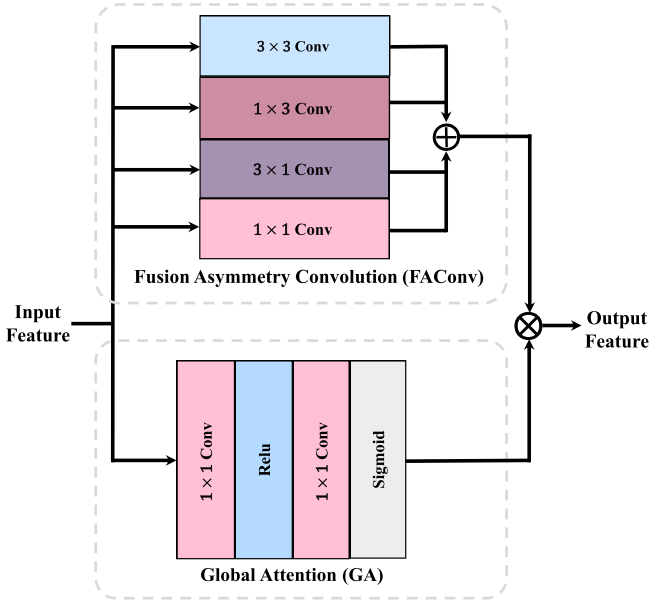


Fig. 2. Architecture of the proposed AAConv, which consists of training phase FAConv and GA. ⊕ denotes the elementwise summation. ⊗ denotes the elementwise product.

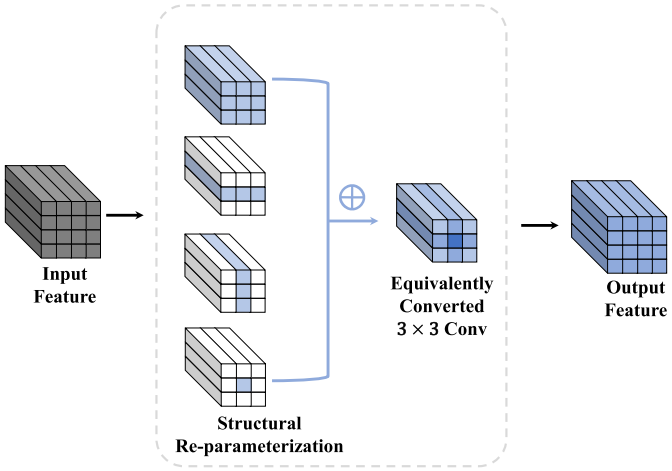


Fig. 3. Deployment phase FAConv with structural reparameterization. ⊕ denotes the elementwise summation.

in a new 3×3 convolution that incorporates the rich spatial and spectral contextual information extracted by the training phase FAConv. This 3×3 convolution, which fuses the parameters of four parallel convolutions, inherits the superior performance of the training phase FAConv. Below we will provide a detailed explanation of how to convert the multibranch FAConv into a 3 × 3 convolution during model deployment.

The dth convolutional kernel is denoted as $\mathcal{K}^{(d)} \in \mathbb{R}^{K\times K\times C}$ ($d = 1, 2, \ldots, D$), where D represents the number of convolutional kernels, K represents the size of convolution kernel. The process of the 3 × 3 convolutional layer can be represented as

$$\mathbf{O}_{:,:,d} = \sum_{c=1}^{C} \mathbf{F}_{:,:,c} \ * \ \mathcal{K}^{(d)}_{:,:,c} \tag{3}$$

where $*$ represents the 2-D convolution operation, c and d are the indices of C and D, respectively, and $\mathbf{O}_{:,:,d}$ represents the feature map output by the dth convolution kernel $\mathcal{K}^{(d)}_{:,:,c}$ convolved with the intermediate feature map $\mathbf{F}_{:,:,c}$. For simplicity, we have omitted the bias term in the formula. The dth convolutional kernel of the 1 × 3 convolutional layer in the model is denoted as $\mathcal{K}^{v(d)} \in \mathbb{R}^{1\times 3\times C}$, the dth convolutional kernel of the 3 × 1 convolution is denoted as $\mathcal{K}^{h(d)} \in \mathbb{R}^{3\times 1\times C}$, and the dth convolutional kernel of the 1 × 1 convolution is denoted as $\mathcal{K}^{p(d)} \in \mathbb{R}^{1\times 1\times C}$. The entire convolutional process of these three types of convolutions can be, respectively, represented as

$$\mathbf{O}^{v}_{:,:,d} = \sum_{c=1}^{C} \mathbf{F}_{:,:,c} \ * \ \mathcal{K}^{v(d)}_{:,:,c} \tag{4}$$

$$\mathbf{O}^{h}_{:,:,d} = \sum_{c=1}^{C} \mathbf{F}_{:,:,c} \ * \ \mathcal{K}^{h(d)}_{:,:,c} \tag{5}$$

$$\mathbf{O}^{p}_{:,:,d} = \sum_{c=1}^{C} \mathbf{F}_{:,:,c} \ * \ \mathcal{K}^{p(d)}_{:,:,c} \tag{6}$$

where $\mathbf{O}^{v}_{:,:,d}$, $\mathbf{O}^{h}_{:,:,d}$, and $\mathbf{O}^{p}_{:,:,d}$ represent the output feature maps obtained by performing 2-D convolution operations between $\mathcal{K}^{v(d)}$, $\mathcal{K}^{h(d)}$, $\mathcal{K}^{p(d)}$ and $\mathbf{F}_{:,:,c}$. As exhibited in Fig. 3, a deployment phase FAConv can be replaced by an equivalent 3 × 3 convolution

$$\mathbf{O}'_{:,:,d} = \mathbf{O}_{:,:,d} + \mathbf{O}^{v}_{:,:,d} + \mathbf{O}^{h}_{:,:,d} + \mathbf{O}^{p}_{:,:,d} \tag{7}$$

$$\mathcal{K}'^{(d)}_{:,:,c} = \mathcal{K}^{(d)}_{:,:,c} + \mathcal{K}^{v(d)}_{:,:,c} + \mathcal{K}^{h(d)}_{:,:,c} + \mathcal{K}^{p(d)}_{:,:,c} \tag{8}$$

$$\mathbf{O}'_{:,:,d} = \sum_{c=1}^{C} \mathbf{F}_{:,:,c} * \ \mathcal{K}'^{(d)}_{:,:,c} \tag{9}$$

where $\mathcal{K}'^{(d)}_{:,:,c}$ is the dth 3 × 3 convolution kernel obtained after equivalent transformation and $\mathbf{O}'_{:,:,d}$ is the feature map output by performing a 2-D convolution operation between $\mathcal{K}'^{(d)}_{:,:,c}$ and $\mathbf{F}_{:,:,c}$.

The multibranch convolutional structure can use different convolutional kernels to extract different spatial and spectral features from the input data, thereby enhancing the network's representational power. Each branch is then added and merged to generate more comprehensive and complex spatial and spectral feature representations. After converting all FAConv modules during deployment, the equivalently converted 3 × 3 convolution captures the same rich spatial and spectral contextual information as the multibranch structure used during training. Additionally, the converted convolution maintains the same level of efficiency as the standard 3 × 3 convolution during deployment.

To make AAConv pay more attention to the pixels in HSIs that are affected by haze pollution without increasing the model's computational load and the number of parameters, we designed a simple GA structure consisting of two 1 × 1 convolutional layers, Relu activation function, and Sigmoid activation function in series. Although GA does not significantly increase the model's computational load, it can significantly improve its performance. AAConv, as the main component, gradually extracts advanced spatial patterns and

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spectral features, while adaptively adjusting the weights of each pixel to focus more on the areas affected by haze.

C. PCSA

To improve the accuracy of restoring spectral information in regions affected by haze, we developed the PCSA module as shown in Fig. 1. The module is proposed to self-learn the correlations between different channels and finely adjust the weights of each channel. 2-D average pooling is designed to transform the global spatial information into the channel descriptor, which can be represented as

$$p_c = \frac{1}{H \times W} \sum_{i}^{H} \sum_{j}^{W} \mathbf{F}_p^{i,j,c} \tag{10}$$

where $\mathbf{F}_p^{i,j,c}$ represents the value of the cth channel of feature $\mathbf{F}_p$ at coordinate (i, j), p_c represents the channel descriptor. 2-D average pooling reduces the size of the input feature map from $H \times W \times c$ to $1 \times 1 \times c$.

To finely adjust the weights of each channel, a linear transformation is applied to the pooled feature map, followed by computing the self-attention matrix for channel vectors. The process of computing the self-attention matrix $\mathbf{A}$ can be represented as follows:

$$Q = \mathbf{W}^q p_c, \quad K = \mathbf{W}^k p_c, \ \mathbf{A} = K^T \times Q \tag{11}$$

where Q and K represent the vectors obtained after linear transformations $\mathbf{W}^q$ and $\mathbf{W}^k$.

After performing 1-D average pooling on $\mathbf{A}$ to obtain a vector, it is input into a 1-D convolution and passed through a Sigmoid activation function to obtain the finely-tuned channel weights. Finally, the channel weights are multiplied with the corresponding elements of the input features to obtain the final output of PCSA. The process of 1-D pooling can be represented as follows:

$$\mathbf{F}_p^* = (\sigma(1\text{DConv}(v_c))) \times \mathbf{F}_p \tag{12}$$

where v_c represents the vector obtained after 1-D pooling, σ is the Sigmoid function, 1DConv is 1-D convolution, and $\mathbf{F}_p^*$ is the final output of PCSA.

IV. Experiments

A. Datasets

In this study, we evaluate the performance of our proposed method using both simulated and real datasets. Specifically, we leverage Chinese Gaofen-5 satellite data to construct a baseline simulated haze hyperspectral dataset. Additionally, we generate a real haze hyperspectral dataset by utilizing the Earth Observing-1 (EO-1) satellite data obtained from the United States Geological Survey (USGS). Detailed descriptions of these datasets are provided below.

1) Simulated Dataset: To quantitatively evaluate the dehazing performance, a simulated haze dataset is required due to the unavailable ground-truth data after haze removal under realistic conditions. We use the cloud simulation method in [44] to generate the simulated hazy HSIs. First, we randomly selected the cirrus band from a Landsat-8 image as the haze intensity map. Next, the spectral vector of real haze from a high-resolution Gaofen-5 image was extracted. Then, we multiplied the haze intensity map with the spectral vector of haze to obtain a pure haze image. Finally, we added the pure haze image to the haze-free HSI to obtain the haze-contaminated HSI. During the fusion process, we used a small coefficient to adjust the thickness of the haze. Finally, a total of 210 clear Gaofen-5 images with a size of $256 \times 256 \times 150$ were employed as the ground truth. The corresponding simulated hazy images were produced by overlaying a haze patch from the hazy HSI, in which 170 were used for training, 20 each were set aside for validation and testing purposes. Please refer to [44] for more details of the simulation process.

2) Real Dataset: To implement AACNet on the real remote sensing data, we built a dataset utilizing EO-1 Hyperion images, which have 242 continuous spectral bands with wavelengths from 0.35 to 2.58 μm and a spatial resolution of 30 m. We collected 43 pairs of hazy and clear HSIs, where the clear images were chosen at the same location as the corresponding hazy image, but captured on the nearest dates. Out of the 43 pairs, 37 pairs were used as the training samples and six pairs were test samples. Each HSI in the dataset was cropped to a size of $256 \times 256 \times 242$. Both the simulated dataset and the real dataset include various land cover types, such as farmland, rivers, mountains, cities, plains, and lakes, among others. This ensures sufficient heterogeneity in the dataset, allowing us to evaluate the generalization performance of our proposed model.

B. Implementation Settings

The paired hazy and clear HSIs in simulated and real datasets are all cropped to a spatial size of 64×64 for training and testing. The training batch size is set as 32. The proposed AACNet model is optimized using the adaptive moment estimation (ADAM) [45] optimizer, where $beta_1 = 0.9$, $beta_2 = 0.99$ and epsilon $= 10^{-8}$. The initial learning rate is set to 0.0002, and the training stops after 50 epochs. We found experimentally that the model works best when the number of RAABs is 5, so we set $n = 5$. All experiments are completed on the PyTorch framework via an NVIDIA A40 and Intel[1] Xeon[1] Silver 4314 CPU. In this article, we optimize the network by adopting the mean-squared error (mse) as the loss function to measure hazy and clear HSIs.

C. Competing Methods

To evaluate the effectiveness of the proposed dehazing method, we compared it against several state-of-the-art dehazing methods in both computer vision and remote sensing domains, including multiscale CNNs (MSCNNs) [12], gated context aggregation network (GCANet) [14], SGNet [29], all-in-one network (AODNet) [46], physics-based feature dehazing networks (PFDNs) [47], multiinput attention network (MIAN) [48], aerial image dehazing transformer (AIDTransformer) [49], and a variant network of U-Net [24] for dehazing (GUNET_S) [16]. These dehazing methods were selected as

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TABLE I
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE SIMULATED VALIDATION AND TEST DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Models	Params(M)	Flops(G)	Validation				Test			
			RMSE (↓)	UQI (↑)	SAM (↓)	SSIM (↑)	RMSE (↓)	UQI (↑)	SAM (↓)	SSIM (↑)
MSCNN	0.13	8.24	0.0379	0.6718	11.04	0.9108	0.0361	0.6911	10.27	0.9172
GCANet	0.79	24.58	0.0308	0.7984	4.62	0.9680	0.0287	0.8084	4.38	0.9656
SGNet	0.46	52.3	0.0149	0.9490	1.78	0.9802	0.0152	0.9425	2.02	0.9797
AODNet	4.36	286.15	0.0131	0.9421	2.02	0 9885	0.0115	0.9464	1.94	0.9892
PFDN	11.44	1.57	0.0106	0.9640	1.73	0.9824	0.0105	0.9614	1.77	0.9822
MIAN	200.94	109.25	0.0101	0.9499	1.77	0.9900	0.0095	0.9459	1.78	0.9903
AIDTransformer	10.90	203.23	0.0083	0.9627	1.58	0.9911	0.0088	0.9672	1.57	0.9911
GUNet_S	2.67	14.11	0.0035	0.9942	0.61	0.9983	0.0042	**0.9944**	0.65	0.9978
Ours	1.82	110.43	**0.0029**	**0.9952**	**0.56**	**0.9986**	**0.0033**	**0.9944**	**0.61**	**0.9982**

our competitors based on their high citation count. All parameters were set according to the author's recommendations in their papers. MSCNN, GCANet, and AODNet are three classic CNN-based dehazing networks for natural images. SGNet divides each HSI into multiple spectral subsets and inputs them to parallel branch convolutions for feature extraction to obtain the dehazed HSI. PFDN employs an encoder–decoder architecture with residual learning and incorporates a physical model unit that simulates the haze formation process to dehaze natural images. MIAN is a multiinput CNN based on the encoder–decoder architecture designed for multispectral remote sensing image dehazing. AIDTransformer is an aerial image dehazing network based on the vision Transformer [50] architecture. GUNET_S replaces the convolution blocks in U-Net with residual blocks and introduces attention mechanisms to improve the network's representation capability, and is a recent dehazing network designed for natural images.

D. Experiments on the Simulated Dataset

1) Evaluative Metrics: Four widely used evaluation metrics, including root mse (RMSE), universal quality index (UQI) [51], spectral angle mapper (SAM) [52], and structural similarity index measure (SSIM) [53] were adopted

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(H_i - \hat{H}_i\right)^2} \tag{13}$$

$$\mathrm{UQI} = \frac{4\sigma_{xy}\mu_x\mu_y}{\left(\sigma_x^2+\sigma_y^2\right)\left(\mu_x^2+\mu_y^2\right)} \tag{14}$$

$$\mathrm{SAM} = \cos^{-1}\left(\frac{\mathbf{t}\cdot\mathbf{p}}{|\mathbf{t}||\mathbf{p}|}\right) \tag{15}$$

$$\mathrm{SSIM}(x,y) = \frac{\left(2\mu_x\mu_y+c_1\right)\left(2\sigma_{xy}+c_2\right)}{\left(\mu_x^2+\mu_y^2+c_1\right)\left(\sigma_x^2+\sigma_y^2+c_2\right)} \tag{16}$$

where n is the number of pixels in the image, H_i is the true value for pixel i, $\hat{H}_i$ is the predicted value for pixel i. σ_{xy} is the covariance between the two images, σ_x^2 and σ_y^2 are the variances of the two images, and μ_x and μ_y are the means of the two images. In SAM computation, $\cos^{-1}$ is the inverse cosine function, $\mathbf{t}$ is the spectral signature of the true spectrum, $\mathbf{p}$ is the spectral signature of the predicted spectrum. In SSIM computation, c_1 and c_2 are constants to avoid instability when the denominator is close to zero.

RMSE reflects the average absolute error between the reconstructed image and the ground truth image, with lower values indicating better reconstruction quality. UQI measures the similarity in terms of luminance, contrast, and structure. As the UQI value increases, the image quality is expected to improve accordingly. SAM measures the spectral similarity. A lower SAM value indicates a higher spectral similarity between the reconstructed image and the ground truth image. SSIM calculates the structural similarities, with higher values indicating better reconstruction quality. It is worth noting that all metrics were computed by taking an average over all channels.

2) Quantitative Results: Table I shows the quantitative results of all comparison methods on the simulated validation and test dataset. The results presented in Table I indicate that our method consistently achieves superior performance over all other dehazing methods in the validation dataset across all metrics. In the test dataset, our method demonstrates comparable performance to GUNET_S in terms of the UQI metric, while outperforming other dehazing methods in other metrics. Compared to the representative MSCNN method, AACNet reduces RMSE from 0.0379 to 0.0029, increases UQI from 0.6718 to 0.9952, decreases SAM from 11.4 to 0.56, and increases SSIM from 0.9108 to 0.9986 on the validation dataset. Similarly, on the test dataset, AACNet reduces RMSE from 0.0361 to 0.0033, increases UQI from 0.6911 to 0.9944, decreases SAM from 10.27 to 0.61, and increases SSIM from 0.9172 to 0.9982. Compared to the latest GUNET_S method, AACNet reduces RMSE by 0.0006, increases UQI by 0.001, decreases SAM by 0.05, and increases SSIM by 0.0003 on the validation dataset. On the test dataset, AACNet reduces RMSE by 0.0009, decreases SAM by 0.004, and increases SSIM by 0.0004. The parameters of AACNet decreased by 0.85M and 9.08M compared to GUNet_S and AIDTransformer, respectively. The floating-point operands of AACNet decreased by 92.8G compared to AIDTransformer. Overall, our proposed AACNet achieves a balance between dehazing performance and model complexity.

3) Qualitative Results: For visual evaluation, we randomly selected six different scenes with varying levels of complexity

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Fig. 4. Haze removal results of six scenes obtained by different methods. The first and second rows are the images contaminated by haze and their corresponding clear images. The third to 11th rows show true-color composite images of Band 20 (472 nm), Band 35 (536 nm), and Band 65 (664 nm) of the HSI after dehazing.

from the validation and test datasets, such as mountains, lakes, urban, and farmlands. To investigate the visual results of different methods, we selected Band 20 (472 nm), Band 35 (536 nm), and Band 65 (664 nm) from the HSI to composite a true-color image as shown in Fig. 4. It can be seen that MSCNN and GCANet can reduce the thickness of haze in

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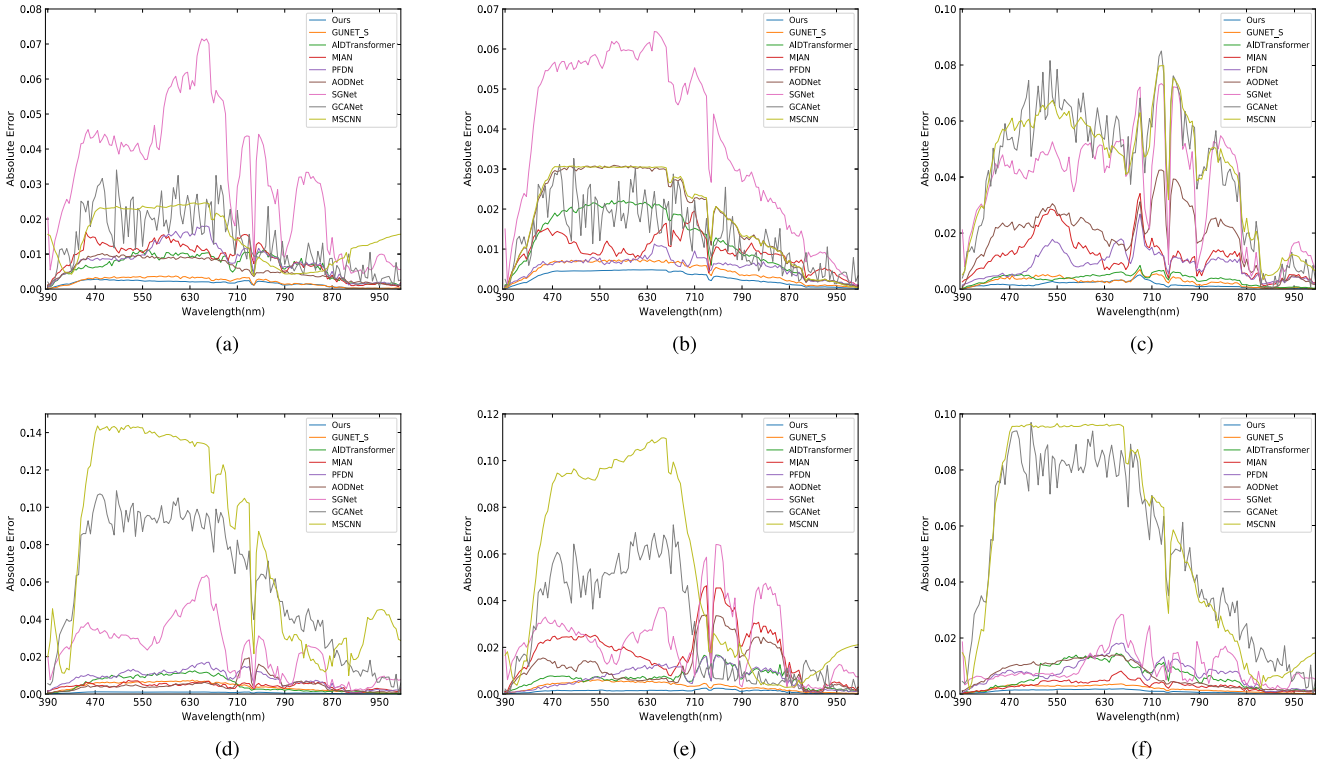


Fig. 5. Absolute error curve of the spectral response corresponds to the six scenes in Fig. 4. The absolute error curve of the spectral response is obtained by calculating the absolute error of the average spectral reflectance of hazy areas with a spatial size of 5 × 5 pixels. (a) Scene 1. (b) Scene 2. (c) Scene 3. (d) Scene 4. (e) Scene 5. (f) Scene 6.

hazy regions, but a significant amount of haze was still retained in the image. In scene 2, SGNet, AIDTrasformer and GUNet_S showed obvious color distortion, and AODNet had haze remaining in the removed image. In scenes 3 and 4, SGNet, PFDN, and MIAN exhibited color distortion, and AODNet had a small amount of haze remaining. In scene 5, SGNet, AODNet, PFDN, MIAN, and AIDTrasformer all show obvious color distortion. In scene 6, SGNet, AODNet, and PFDN showed brighter colors and were not as clear as the reference image in the marked area (highlighted by a red box). Overall, the haze-free images recovered by our proposed AACNet method had the best visual results and were closest to the reference image.

To further compare the spectral reconstruction performance of different methods, we calculated the absolute error between the average spectral reflectance of regions with a spatial size of 5 × 5 pixels in different HSIs and the spectral reflectance of the corresponding regions in the reference HSI (see Fig. 5). Our analysis showed that the spectral response curves restored by MSCNN and GCANet differed significantly from the curves of the reference HSI, indicating poor spectral reconstruction performance. The spectral response curves restored by SGNet, AODNet, PFDN, MIAN, AIDTrasformer, and GUNET_S were relatively close to the curves of the reference HSI, but the spectral reconstruction performance was poor in bands with high spectral reflectance. In contrast, our proposed AACNet method restored a spectral response curve that was closer to the curve of the reference HSI, indicating superior spectral reconstruction performance and the ability to effectively reconstruct spectra contaminated by haze pollution.

Fig. 6 shows the absolute error heatmaps of six scenes recovered by different methods to the corresponding ground truth image. The images reconstructed by MSCNN and GCANet methods have significant errors in all scenes, while the HSI reconstructed by SGNet has large errors in scenes 1–5. The HSI reconstructed by PFDN and MIAN has noticeable errors in scenes 2 and 3. The HSI reconstructed by AIDTransformer has noticeable errors in scenes 2–6. The HSI reconstructed by GUNet_S has much larger errors in scene 2 compared to the image reconstructed by our proposed AACNet. Obviously, our analysis demonstrated that the haze-free HSIs reconstructed by AACNet had the smallest error compared to the ground truth image Additionally, our proposed AACNet outperforms other methods in terms of dehazing performance. Overall, the reconstruction results of all methods in Scene 2, as indicated by the heatmaps, are the worst compared with other scenes. This is primarily due to the presence of diverse land types such as trees, farmland, and rivers in Scene 2. Additionally, the texture information in Scene 2 images is complex. These two challenges make the reconstruction even more difficult. However, our model still achieves the best result among all methods in Scene 2.

4) Ablation Analysis: In this section, we demonstrate the impact of AAConv and PCSA on our proposed network through ablation studies on the simulated dataset. The experimental results on the test dataset are shown in Table II, where "Backbone" refers to the baseline network with a 3 × 3 convolution replacing AAConv and PCSA. We recorded the best RMSE and SSIM values.

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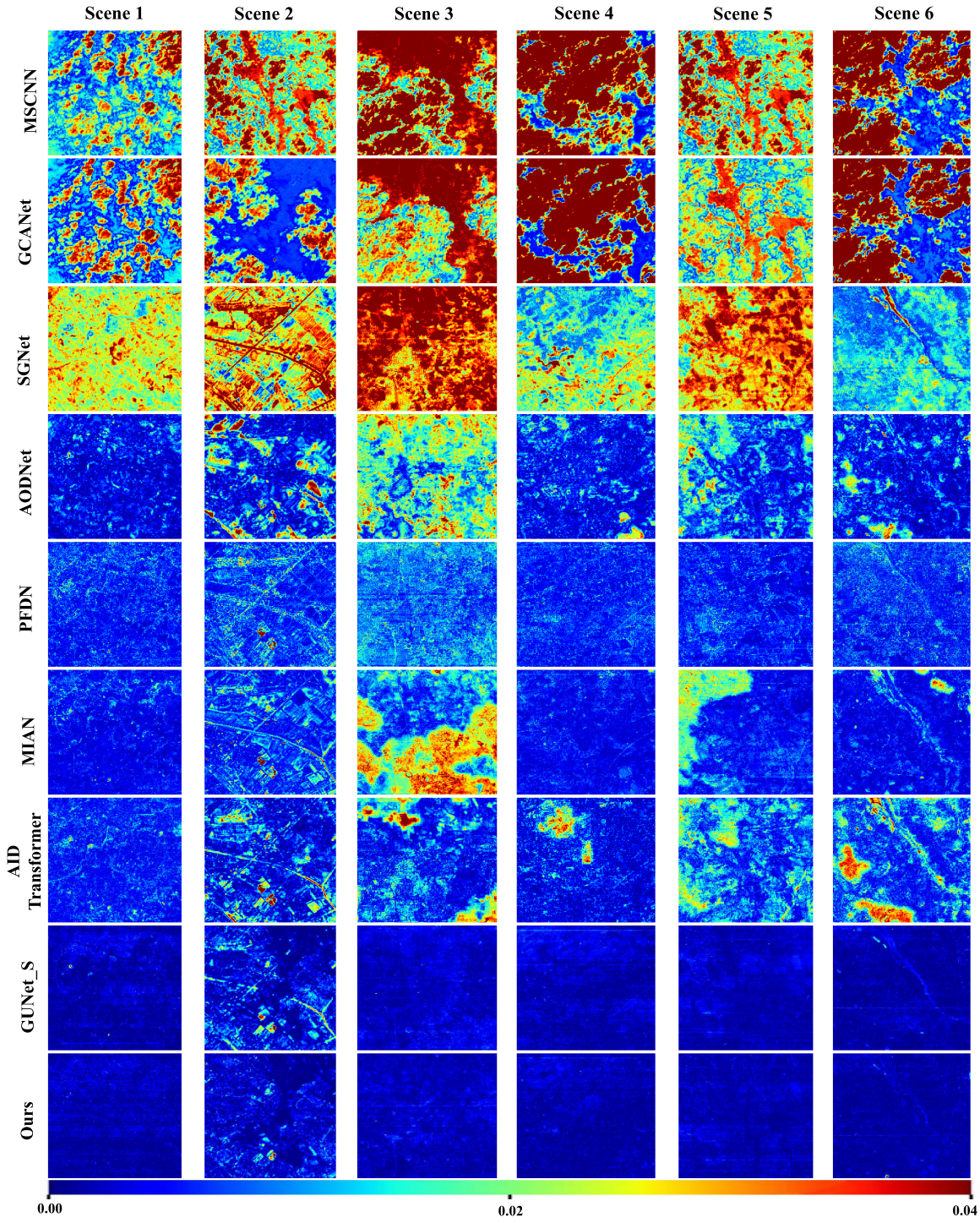


Fig. 6. Average absolute error heatmaps between the restored HSIs using nine different dehazing methods and the corresponding ground truth HSIs.

1) *AAConv:* We investigate the impact of the two submodules of AAConv, FAConv and GA, on the performance of the backbone model. Table II presents the evaluation metrics on the test dataset, where the values of RMSE and SSIM for the backbone network are 0.59 and 0.9974, respectively. We introduce E_a, which represents the model with GA merging into the backbone network, to unequally process different features and pixels, making the backbone network more attentive to the hazy areas and outperforming the baseline. Building on the results of E_a, we further conduct another experiment E_b to examine the impact of FAConv in our proposed model. Compared to the specific network E_a without FAConv, E_b achieves a decrease of 0.0007 in RMSE and an increase of 0.0001 in SSIM. Our experiments demonstrate that the AAConv

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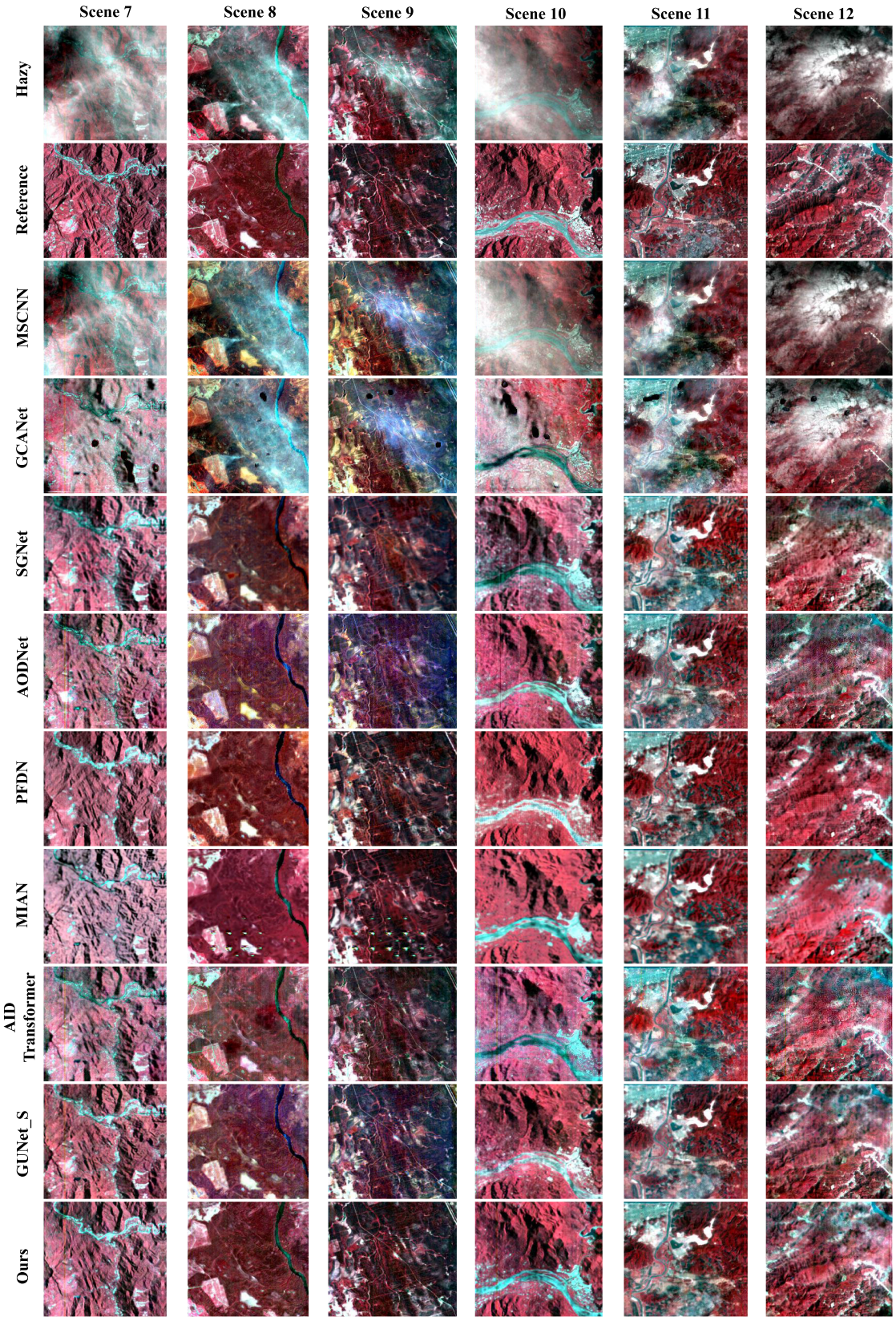


Fig. 7. Haze removal results obtained by different methods on six EO-1 Hyperion image scenes. The first row is the image contaminated by real haze. The second row is the clear reference image collected on the adjacent date. The third to eleventh rows show the false-color composite images after dehazing by nine methods.

composed of GA and FAConv achieves superior performance.

2) *PCSA:* Embedding PCSA into E_a and E_b, respectively, resulted in E_c and E_d. Compared to E_a, E_c exhibits

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TABLE II
ABLATION RESULTS ON THE GAOFEN-5 TEST SET

Description	GA	FAConv	PCSA	RMSE	SSIM
$Backbone$	✗	✗	✗	0.0059	0.9974
E_a	✓	✗	✗	0.0058	0.9975
E_b	✓	✓	✗	0.0051	0.9976
E_c	✓	✗	✓	0.0045	0.9977
E_d	✓	✓	✓	0.0033	0.9982

a decrease of 0.0013 in RMSE and an increase of 0.0002 in SSIM. Similarly, E_d outperforms E_b by achieving a decrease of 0.0018 in RMSE and an increase of 0.0006 in SSIM. These results demonstrate that PCSA contributes to the spectral reconstruction of the hazy area and improves the accuracy of the restored spectra.

E. Experiments on the Real Dataset

The evaluation of haze removal methods on real HSIs is crucial, but obtaining real ground-truth images is challenging, and therefore, visual comparison is generally employed. In Fig. 7, we present the results of seven dehazing methods applied to six different real HSI scenes. The visual results reveal that MSCNN performs similar to its performance on the simulated dataset, and it is ineffective in removing haze in real HSIs. On the other hand, using GCANet for dehazing may result in blurry HSIs with residual haze. SGNet effectively removes haze from most scenes, but a small amount of haze remains in scene 10, and severe color distortion is observed in scenes 7 and 12. AODNet exhibits severe color distortion in scene 8, and there is still haze in scenes 9 and 12. PFDN exhibits severe color distortion in scenes 8, 10, and 12. MIAN exhibits color distortion in scenes 7, 10, and 12, and noticeable noise in scenes 8 and 9. AIDTransformer suffers from severe color distortion in scenes 7 and 10. GUNet_S effectively removes haze in most scenes, but a small amount of haze remains in scenes 8 and 9. In summary, the visual results demonstrate that the comparison methods suffer from issues such as blurring, residual haze, and severe color distortion, making them unsuitable for practical applications.

In contrast, our proposed AACNet effectively removes haze and reconstructs objects contaminated by haze. Compared to other state-of-the-art methods, AACNet can reconstruct the HSI with colors closer to the clear image without causing blurring or residual haze. It can be seen from the simulated and real experimental results, AACNet showed better dehazing results on the crop and mountainous areas, which is mainly because of the simple texture structure in these areas. These results demonstrate that AACNet is an effective method for haze removal in real HSIs, and it can be used to significantly improve the quality of HSIs, especially in scenarios with challenging atmospheric conditions.

It is important to note that while the proposed method delivers promising results in terms of haze removal, it does exhibit slight color distortion in the restored HSIs due to the suboptimal quality of some training data. Moreover, the varying time periods between the hazy and clear images may result in changes in ground objects, causing inconsistent ground objects in the image pairs. This highlights the need for high-quality datasets to improve haze removal results. The presence of complex atmospheric conditions such as multilayered haze and nonuniform haze distribution can also affect the performance of haze removal algorithms. There are other challenges associated with the practical application of haze removal algorithms in real-world scenarios. Nonetheless, the proposed method demonstrates outstanding performance in removing haze from real datasets, effectively restoring the obscured ground objects, and significantly improving the overall quality of HSIs.

V. CONCLUSION

Compared to haze removal in computer vision or multispectral remote sensing fields, haze removal in HSIs presents a greater challenge. Despite this, the number of studies on haze removal in HSIs is limited. To address this gap, this article proposes the AACNet model, which is tailored specifically for HSI haze removal. The model's backbone structure is composed of multiple RAAGs, each containing multiple RAABs, which effectively exploit the spatial and spectral correlations between clear and polluted HSIs affected by haze. During deployment, the trained AACNet can successfully convert all the multibranch FAConvs to a 3×3 convolution while maintaining the powerful feature extraction ability of FAConv. Furthermore, to further improve the haze removal performance of AACNet, we developed a PCSA block to capture the spectral correlations between different spectra of the intermediate features. This block adaptively enhances important spectral features while suppressing redundant features, improving the spectral reconstruction quality of hazy regions. The effectiveness of AACNet for haze removal was validated on both simulated and real hyperspectral datasets, demonstrating superior performance compared to other advanced haze removal methods. Our work highlights the importance of addressing the unique challenges of haze removal in HSIs and provides a promising solution for future research in this area. Overall, the proposed AACNet model showcases the potential for further advancements in haze removal for HSIs. Of course, our work has some limitations. First, the proposed method requires a large amount of graphics processing unit (GPU) memory during training due to the use of multibranch convolution to improve performance and a large amount of data in a single HSI. Second, the quality of training samples can significantly affect the results, which poses a challenge in creating high-quality hyperspectral training data. Third, various noises in HSIs may also affect the haze removal results.

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